

GIANTS CASTLE, South Africa

WMO: 685890

Lat: 29.2650S

Lon: 29.5222E

Elev: 5781

StdP: 11.87

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 32.9	(c) 35.4	(d) -0.7	(e) 6.5	(f) 53.4	(g) 5.3	(h) 9.0	(i) 53.6	(j) 22.8	(k) 53.3	(l) 18.3	(m) 54.5	(n) 4.1	(o) 180	(p) 0.419

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 15.3	(c) 82.6	(d) 58.9	(e) 80.2	(f) 58.7	(g) 77.9	(h) 58.6	(i) 65.7	(j) 75.1	(k) 64.5	(l) 73.4	(m) 63.4	(n) 71.7	(o) 8.9	(p) 20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.8	106.2	69.7	61.6	101.9	68.4	60.6	98.2	67.3	34.1	75.4	33.0	73.6	32.1	72.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 16.7	(b) 14.0	(c) 12.0	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(4) 16.7	(5) 14.0	(6) 12.0	(7) 24.0	(8) 68.2	(9) 2.3	(10) 1.3	(11) 22.4	(12) 69.1	(13) 21.0	(14) 69.9	(15) 19.7	(16) 70.7	(17) 18.1	(18) 71.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.6	64.2	63.4	62.6	58.9	55.4	51.6	50.9	54.9	57.9	59.1	60.4	63.8
	DBStd	7.41	4.58	4.98	5.06	5.57	5.48	4.94	5.65	6.45	8.36	7.35	6.54	5.29
	HDD50	193	0	1	1	6	17	39	58	27	27	12	5	0
	HDD65	2594	71	78	105	193	298	402	437	318	234	209	164	85
	CDD50	3321	439	377	391	272	185	87	86	179	265	295	317	428
	CDD65	247	45	34	30	9	2	0	0	4	22	26	26	49
	CDH74	1668	257	148	134	50	11	2	1	31	224	243	226	341
	CDH80	233	38	12	9	3	0	0	0	1	28	37	37	68
Wind	WSAvg	5.2	4.2	4.1	4.4	4.6	5.3	6.0	6.3	6.1	6.1	5.1	4.9	4.6
Precipitation	PrecAvg	41.1	7.9	6.3	5.7	2.8	0.7	0.5	0.2	1.2	2.0	3.3	4.9	5.5
	PrecMax	54.1	15.6	10.6	12.9	6.0	2.1	3.9	0.9	5.7	5.3	7.2	10.4	10.2
	PrecMin	30.2	1.6	2.9	1.4	1.2	0.0	0.0	0.0	0.0	0.0	1.2	3.0	2.5
	PrecStd	6.7	3.3	2.4	3.0	1.3	0.7	1.0	0.3	1.5	1.7	1.5	1.9	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.2	82.1	81.0	79.4	75.6	71.5	71.1	77.8	83.6	84.1	84.3	85.9
		MCWB	61.9	61.7	61.1	59.6	52.6	46.9	45.4	50.9	53.0	56.1	58.7	61.2
	2%	DB	79.9	78.4	77.7	74.9	71.5	67.6	67.4	73.2	79.7	80.3	79.6	81.0
		MCWB	62.4	62.8	61.4	57.2	50.8	45.8	44.8	49.1	52.1	55.2	58.5	61.5
	5%	DB	76.6	75.3	74.4	71.4	68.3	64.5	64.2	69.8	76.5	76.4	76.1	77.4
		MCWB	62.5	62.7	61.2	56.6	50.0	44.9	44.0	47.4	51.2	54.9	58.6	61.1
	10%	DB	73.3	72.1	71.2	68.1	65.1	61.2	61.0	66.3	72.8	72.5	72.5	74.0
		MCWB	61.6	62.2	60.4	55.8	49.2	44.1	42.9	46.7	50.4	54.4	58.3	61.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.9	66.9	67.0	63.4	56.5	50.6	49.2	53.9	57.9	62.0	63.8	67.2
		MCDB	77.3	75.7	74.7	72.2	67.9	64.0	63.8	71.1	73.5	72.0	75.6	78.6
	2%	WB	65.0	65.4	64.5	61.2	53.8	48.4	47.1	51.5	55.5	59.6	62.1	65.1
		MCDB	74.5	74.0	72.6	70.0	66.0	61.5	60.2	66.9	70.6	71.1	72.8	74.6
	5%	WB	63.8	64.1	62.9	59.2	52.1	46.9	45.6	49.9	53.8	58.1	60.9	63.6
		MCDB	72.2	72.3	70.7	67.2	64.1	60.1	58.8	63.8	68.5	69.2	71.2	72.4
	10%	WB	62.7	62.8	61.2	57.2	50.5	45.4	44.2	48.3	52.4	56.6	59.4	62.2
		MCDB	70.5	69.8	68.6	65.8	62.4	57.7	57.5	61.3	66.4	66.6	69.2	70.6
Mean Daily Temperature Range	5% DB	MDBR	15.3	14.9	15.8	17.6	20.3	21.8	21.4	21.7	21.5	18.2	17.5	16.6
		MCDBR	22.1	20.8	21.5	22.1	23.4	24.8	23.9	24.9	26.7	26.6	25.5	24.1
		MCWBR	10.9	9.9	10.1	9.9	10.9	12.6	11.6	10.5	9.8	10.1	10.7	10.7
	5% WB	MCDBR	18.4	17.4	17.4	18.4	20.9	22.2	21.9	22.5	23.2	21.2	21.7	19.6
		MCWBR	10.2	9.3	9.7	9.4	10.9	12.1	11.3	10.7	10.2	9.9	10.5	10.8
Clear-Sky Solar Irradiance	taub	0.370	0.368	0.352	0.310	0.280	0.261	0.268	0.329	0.376	0.403	0.363	0.366	
	taud	2.365	2.376	2.395	2.483	2.539	2.560	2.523	2.351	2.236	2.204	2.350	2.366	
	Ebn at Noon	308	303	298	295	292	291	293	284	285	289	308	310	
	Edh at Noon	42	40	37	30	26	24	26	34	43	47	42	42	
All-Sky Solar Radiation	RadAvg	1948	1854	1652	1402	1220	1107	1197	1392	1649	1756	1943	1980	
	RadStd	133	154	104	86	64	62	62	84	98	136	131	117	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	+0.59	+1.03	N/A	N/A	-0.61	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page